Experiment 1
Data Acquisition and Data Profiling using BI Tools

Aim

To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to assess data quality, understand data structure, identify anomalies, and prepare data for further analytical processing.

Tools / Software Required

- Microsoft Power BI Desktop
- Sample datasets: Sales_Data.csv, Employee_DB.xlsx, and a sample SQL database

Objective

- To understand the basic concepts of Business Intelligence, data acquisition, and data profiling.
- To acquire data from multiple heterogeneous sources such as CSV, Excel, and SQL Server.
- To import and integrate different datasets into Microsoft Power BI.
- To examine the structure, data types, completeness, uniqueness, and validity of the datasets.
- To examine the structure, data types, completeness, uniqueness, and validity of the datasets.
- To use Power Query Editor's Column Quality, Column Distribution, and Column Profile features for data profiling.
- To document the identified data quality issues and suggest suitable corrective actions before further analysis.

Theory

Business Intelligence (BI)

Business Intelligence refers to a set of technologies, processes, and practices that enable organizations to collect, integrate, analyse, and present business information. BI helps transform raw data into meaningful, actionable insights that support better decision-making across an organization.

Key components of a BI system include:

- Data Sources: Databases (SQL/NoSQL), flat files, cloud stores, and APIs
- ETL Pipeline: Extract, Transform, Load processes that move data from source to destination
- Data Warehouse / Data Lake: Centralized repositories for structured or unstructured data
- BI & Reporting Tool: Microsoft Power BI for data connection, transformation, and visualization
- Analytics Layer: Descriptive, diagnostic, predictive, and prescriptive analytics

Data Acquisition

Data Acquisition is the process of collecting and importing data from various heterogeneous sources into a unified environment for processing. It is the first and foundational step in any BI pipeline. In Power BI, data is acquired through the Get Data option, which supports a wide range of connectors including flat files (CSV, XLSX), relational databases (SQL Server, MySQL), cloud sources (SharePoint, Azure), and web-based sources.

Data Profiling

Data Profiling is the process of examining data from an existing source and collecting statistics, summaries, and metadata about that data. It is a critical quality assurance step that reveals the structure, content, and quality of a dataset before further processing. Data profiling covers the following key dimensions:

- Completeness: Percentage of non-null values in each column.
- Uniqueness: Ratio of distinct to total values — detects duplicate records.
- Validity: Whether values conform to defined rules or expected formats.
- Consistency: Uniformity of values across related datasets.
- Accuracy: Correctness of values relative to real-world facts.
- Timeliness: Currency of data relative to the reporting period.

Data Profiling in Power BI — Power Query Editor

Microsoft Power BI provides built-in data profiling capabilities through the Power Query Editor. Under the View tab, three profiling features can be enabled:

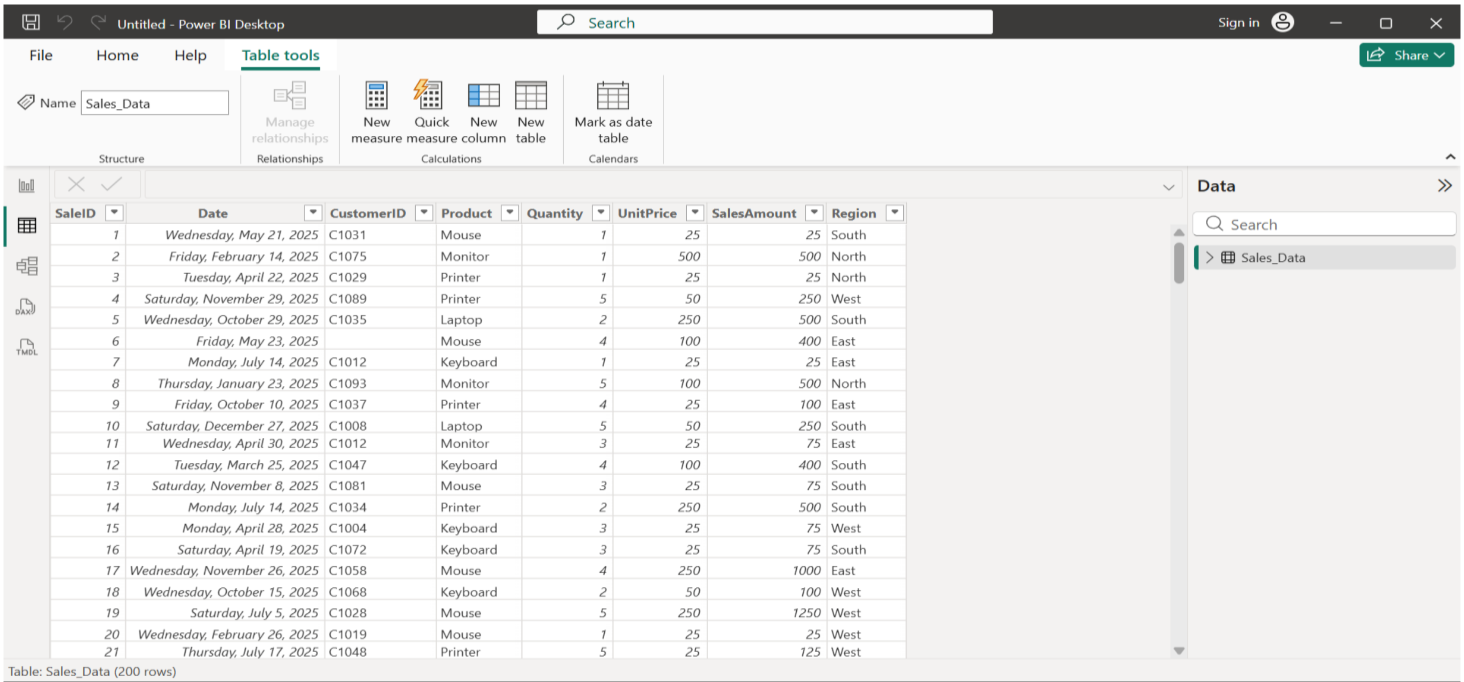
- Column Quality: Displays the percentage of valid, error, and empty values per column.
- Column Distribution: Shows a histogram of value frequency along with the distinct and unique count for each column.
- Column Profile: Provides detailed statistics for a selected column including minimum, maximum, mean, median, standard deviation, and a full value distribution chart.

Conduct

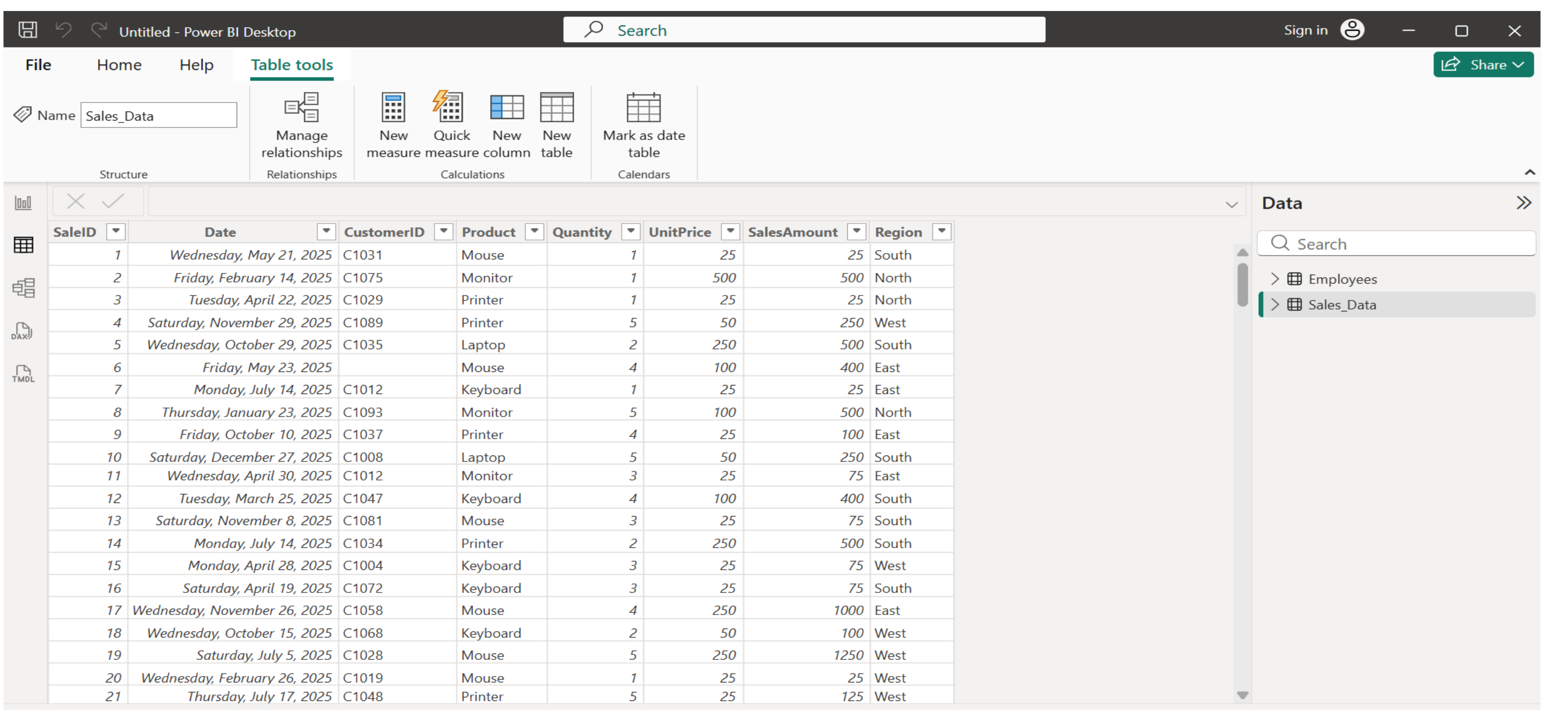
Part A: Data Acquisition

Step 1: Launch Power BI Desktop and import the CSV file.

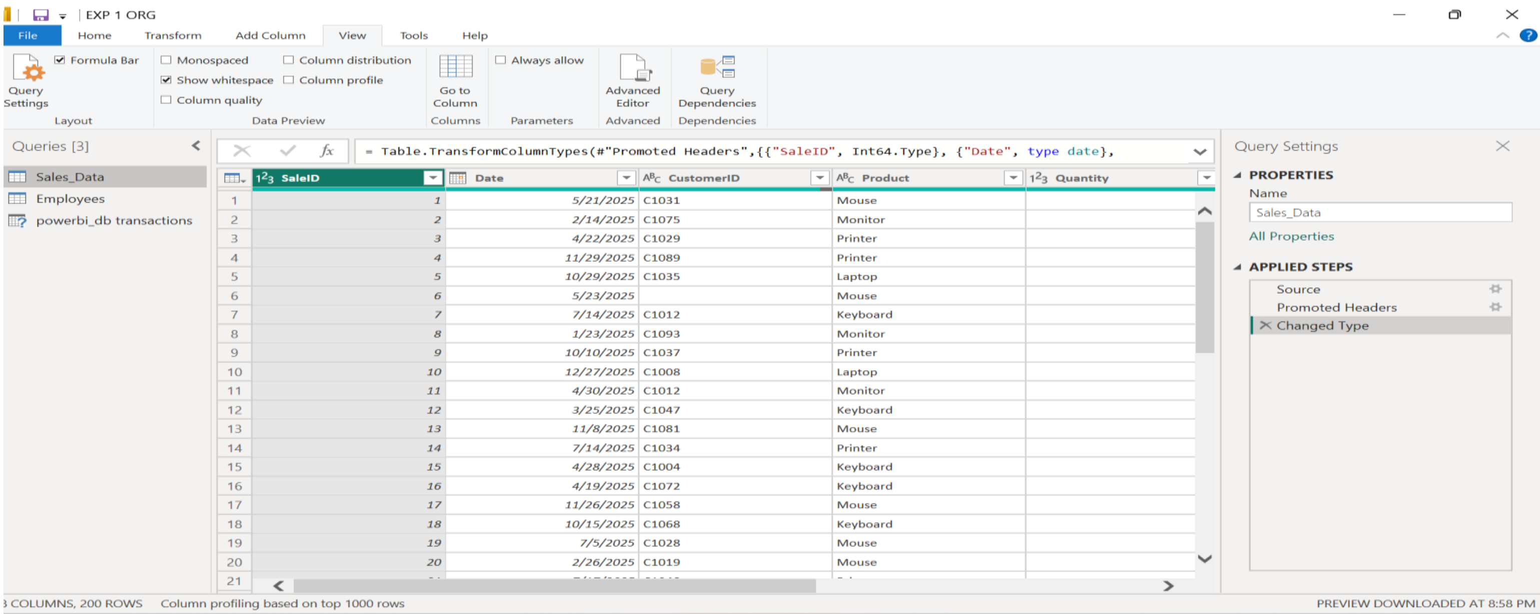
Home > Get Data > Text/CSV
Browse to Sales_Data.csv > Load



Step 2: Import the Excel file.



Home > Get Data > Excel Workbook
Browse to Employee_DB.xlsx > Select the appropriate sheet > Load



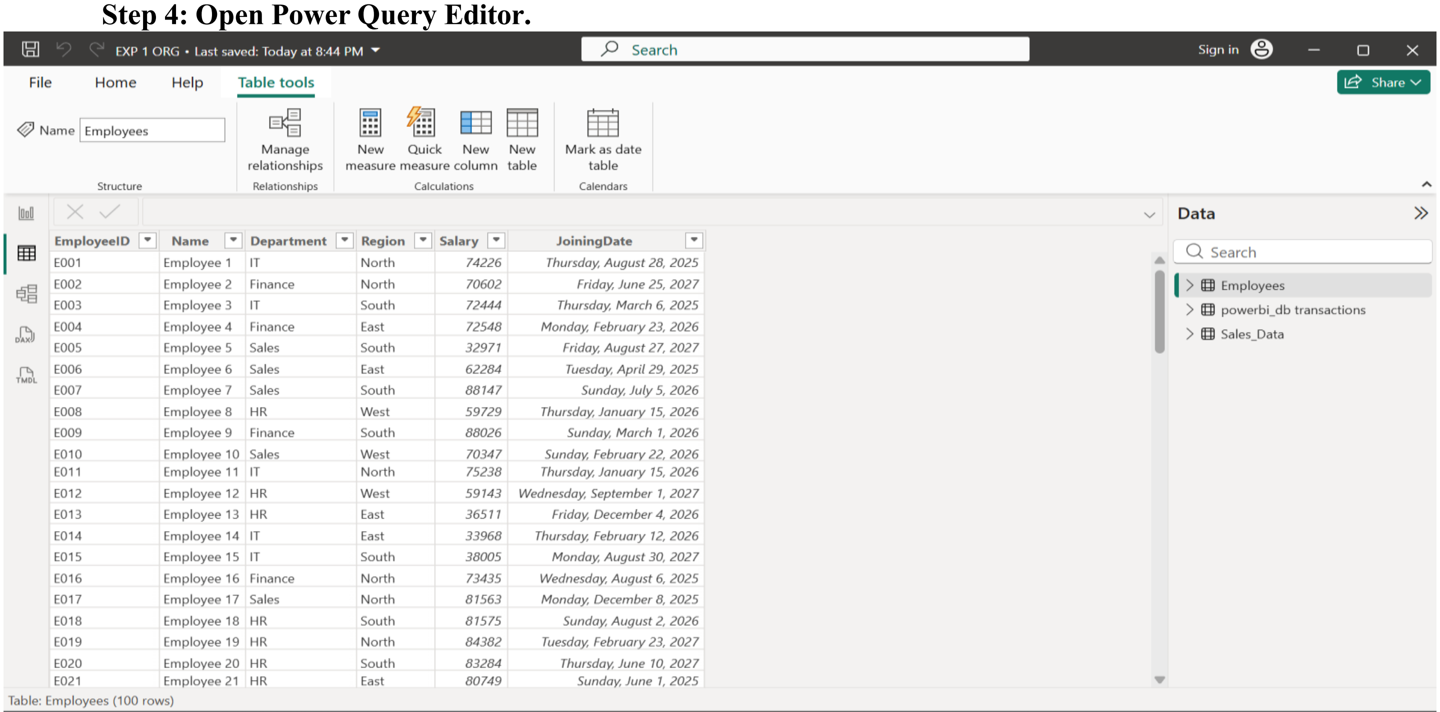
Step 3: Connect to the SQL database and import the Transactions table.

Home > Get Data > SQL Server
Enter server name and database credentials > Select Transactions table > Load

Verify that all three data sources appear in the Data pane on the right-hand side of the Power BI window.

Part B: Data Profiling in Power Query Editor

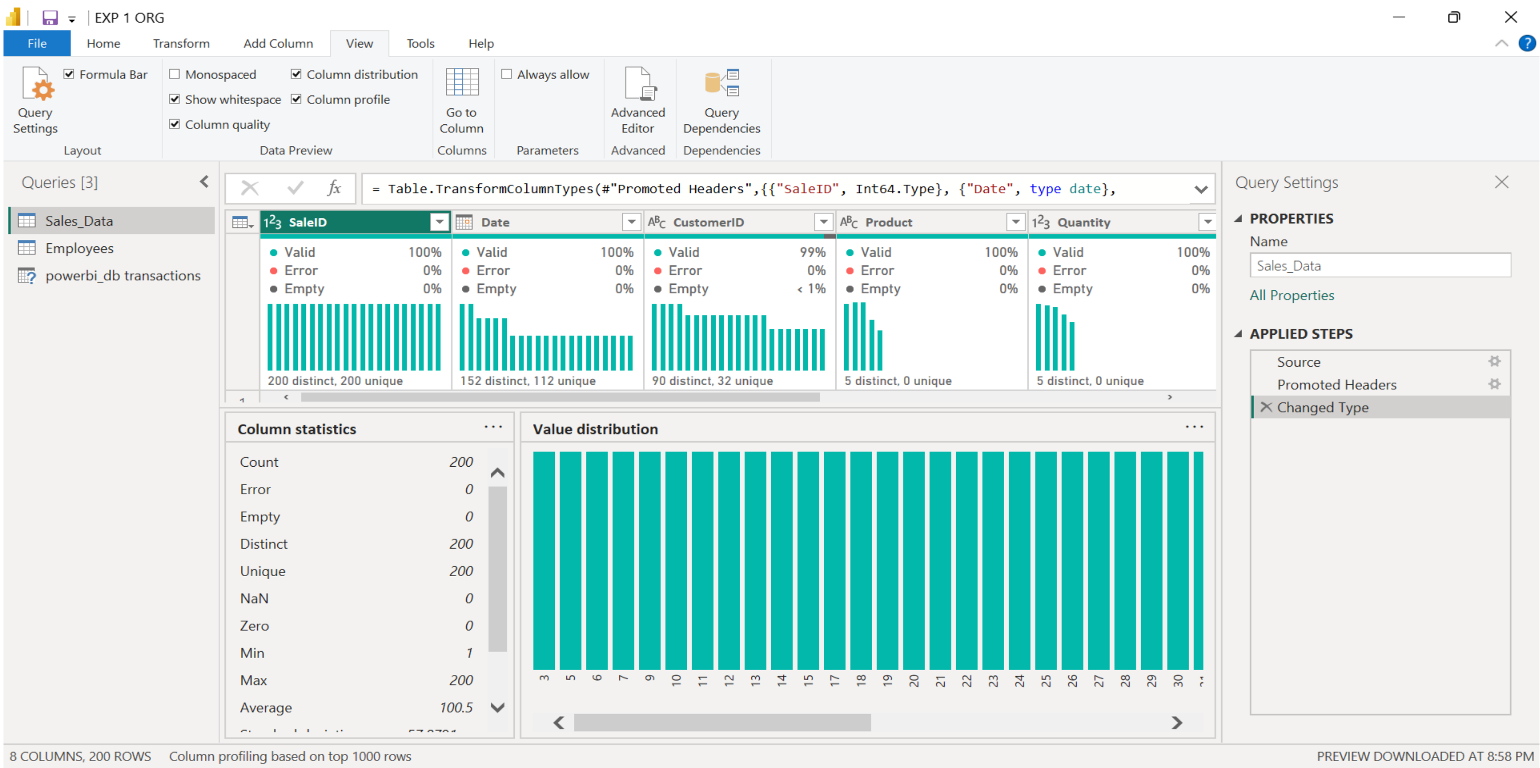
Step 4: Open Power Query Editor.

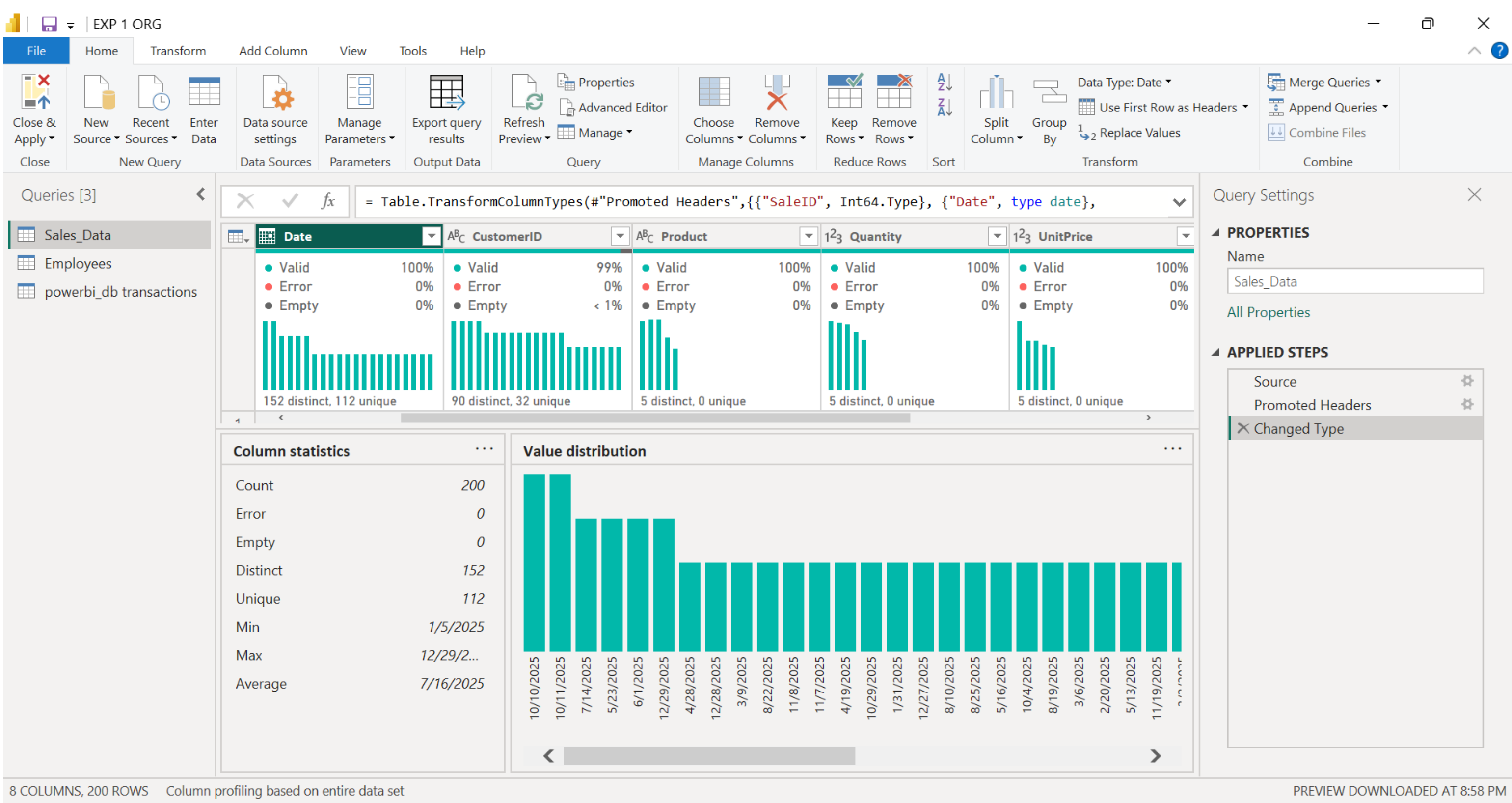


Home > Transform Data

Step 5: Enable data profiling features.

Select each column one at a time and observe the profiling statistics displayed below the data preview grid.

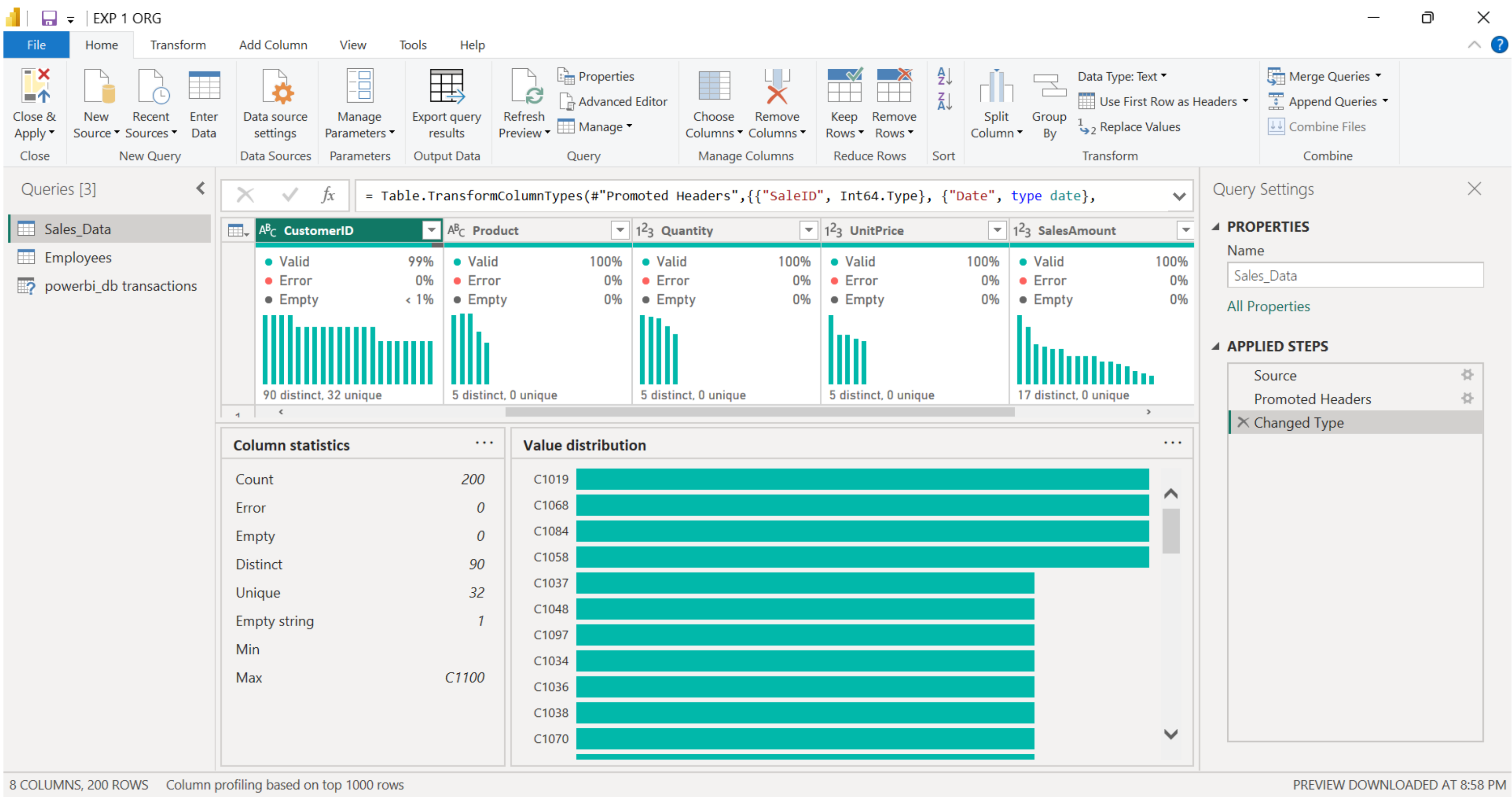




Step 6: Record profiling observations.

For each column in each dataset, note the following:

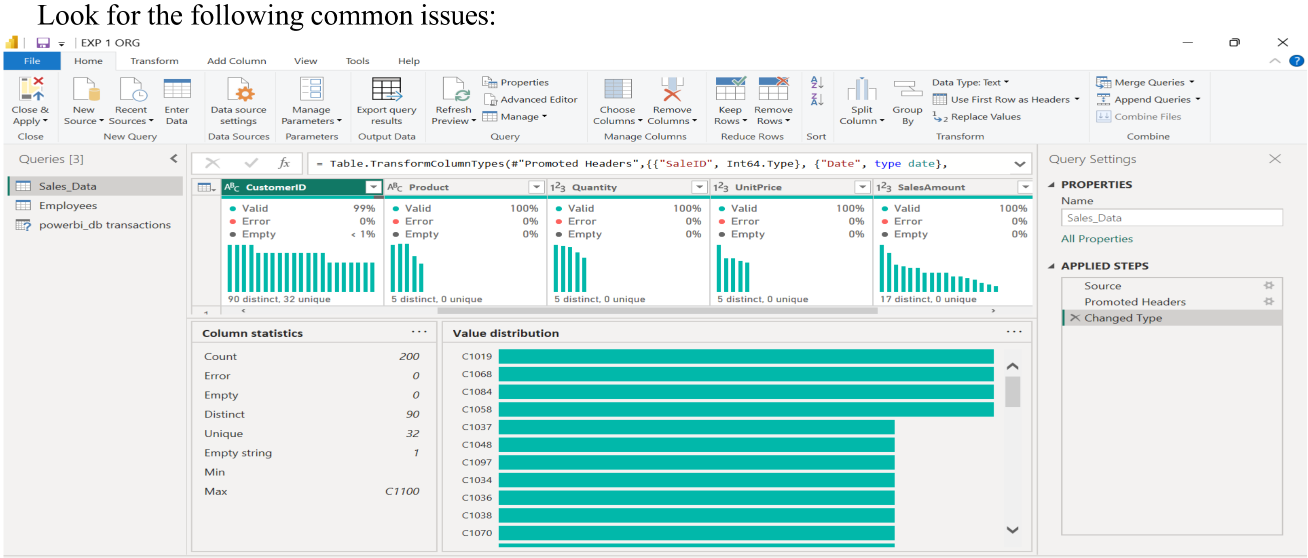
- Data type and total row count
- Null / empty percentage (from Column Quality)
- Distinct value count and unique value count (from Column Distribution)
- Minimum, maximum, mean, and standard deviation for numeric columns (from Column Profile)
- Any errors, mismatched data types, or unexpected values observed



Part C: Identify and Document Data Quality Issues

Step 7: Based on the profiling results, identify data quality issues.

Look for the following common issues:



- High null percentage in important columns such as customer ID or date fields
- Columns with only one unique value — zero variance and no analytical value
- Numeric columns containing text values indicating a data type mismatch
- Date columns with future dates or dates outside the expected range
- Duplicate rows identified through a low uniqueness ratio

Document at least three data quality issues found and note the suggested remediation action for each.

Result

Data from multiple heterogeneous sources (CSV, Excel, and SQL database) was successfully acquired and imported into Microsoft Power BI Desktop. Data profiling was performed using the Power Query Editor's built-in tools — Column Quality, Column Distribution, and Column Profile. Key data quality dimensions including completeness, uniqueness, and validity were assessed across all datasets, and data quality issues were identified and documented for remediation prior to analysis.

Learning Outcomes

After completing this experiment, students will be able to:

- Explain Understood the fundamental concepts of Business Intelligence and data acquisition.
- Gained practical knowledge of acquiring data from CSV, Excel, and SQL Server sources.
- Learned how to import and integrate heterogeneous datasets into Microsoft Power BI.
- Gained hands-on experience with Microsoft Power BI Desktop and Power Query Editor.
- Learned how to use Column Quality to evaluate valid, empty, and error values in datasets.
- Learned how to use Column Distribution to examine distinct and unique values and their frequency.
- Learned how to use Column Profile to examine detailed statistics and characteristics of individual columns.
- Developed the ability to identify missing values, duplicate records, incorrect data types, and inconsistent or unexpected values.
- Understood the importance of data profiling for evaluating data quality before analysis and visualization.
- Gained practical experience in documenting data quality issues and suggesting appropriate remediation actions.
- Developed a better understanding of how clean and reliable data supports accurate business analysis and decision-making.